FINAL YEAR PROJECT REPORT

AI-ENABLED TELEHEALTH APPOINTMENT TRIAGE & PATIENT SCHEDULING COORDINATION SYSTEM

By

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A report submitted in partial fulfilment of the requirements for the award of

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Month Year

DECLARATION

I declare that this Final Year Project report is my own work and that all sources of information have been properly acknowledged. This report has not been submitted for any other degree or professional qualification.

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Date: _____

APPROVAL FOR SUBMISSION

This is to certify that this project report entitled **AI-Enabled Telehealth Appointment Triage & Patient Scheduling Coordination System** was prepared by STUDENT NAME (STUDENT ID) and submitted in partial fulfilment of the requirements for the Bachelor of Information Technology (Hons) at SEGi University.

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ABSTRACT

The proposed solution is an intelligent and lightweight web-based application to fill the structural gap between patient intake and administrative calendar coordination in telehealth environments. Patient intake sorting and the "first-come, first-served" allocation of slots into the virtual clinic are major headaches for modern virtual clinics. The proposed system integrates Natural Language Processing (NLP) text classification with an urgency-aware heuristic scheduling engine. The backend automatically analyzes free-text symptom descriptions entered by the patient and assigns the symptom to predefined medical triage levels (Routine, Urgent, Emergency), which immediately determines the priority of appointment booking. The system creates real-time queue dashboards and an automated scheduling matrix, making queue planning more convenient for clinic coordinators. The main purpose of this study is to assess the capacity of a local, unified AI-based web platform to eradicate the administrative barriers of manual data entry, optimize specialists' time, and reduce clinical waiting times for high-urgency remote consultations in a secure, reliable manner.

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Chapter 1

INTRODUCTION

1.1 Background of the Study

The ongoing transformation of digital healthcare delivery has moved operational friction from local physical clinics to remote web platforms. Telehealth networks have grown exponentially in numbers and have provided unparalleled access to medical care remotely. This shift, however, has turned administrative waiting columns into the primary healthcare constraint, effectively bypassing physical queues at clinics and hospitals only to replace them with digital bottlenecks.

A patient who seeks a virtual consultation will typically provide a simple, written report detailing all of their current symptoms. In normal operational workflows, medical clerks or triage nurses are required to open each digital request, read through unstructured paragraphs of text, manually determine the clinical severity of each case, cross-reference the available doctors' availability charts one by one, and finally schedule an appointment. Major processing latencies are inherently added to this manual processing pipeline, a flaw that is particularly exposed and exacerbated during peak operating times.

The swift emergence of telehealth platforms into the mainstream of health care has undoubtedly helped to democratize patient access, but at the same time, it has revealed significant administrative inefficiencies in virtual clinical workflows. Eliminating physical waiting rooms has solved the problem of requiring patients to be physically present; however, the fundamental administrative mechanics of triage and scheduling are still largely human-driven, manual, and error-prone. This project proposes a web application that allows virtual clinics to automate this intake-to-scheduling transition and helps streamline coordination without incurring the massive resource overhead typically associated with

enterprise-level hospital networks.

1.2 Problem Statement

Healthcare environments today separate patient intake and appointment scheduling—two inherently linked activities—into distinct and disconnected silos. Traditional booking systems approach all consultation requests in exactly the same way, operating on a chronological queue-based system that is prioritized based on submission time rather than the actual clinical need or physical state of the patient. This systemic design introduces three acute problems:

First, it creates dangerous delays for critical patients. Requests are processed in the strict order that they are received. Consequently, a patient experiencing acute, potentially life-threatening symptoms might end up waiting in line behind dozens of routine prescription requests. Due to the blind nature of these intake queues, a person who has an urgent, high-priority symptom may be waiting behind a person seeking answers to basic administrative questions.

Second, the reliance on manual triage generates significant administrative overhead and introduces a high risk of human error. During times of high patient flow, manual triage requires immense administrative effort and frequently introduces inconsistencies in patient severity assessment. This forces healthcare workers to spend a significant portion of their day merely managing and sorting unstructured data, which is a leading cause of administrative staff burnout in modern clinics.

Third, the primary alternatives available to smaller clinics involve outsourced enterprise complexities. Many current automated clinical systems utilize black-box, paid external Large Language Model (LLM) integrations. These outsourced services introduce severe data privacy vulnerabilities regarding Protected Health Information (PHI), incur high runtime operational expenditures, and present deployment challenges that make them highly impractical for small to medium-sized localized medical practices.

There is a clear and pressing need for a localized, lightweight web prototype that can

naturally ingest patient symptom narratives securely, safely calculate an objective clinical priority score using local intelligence, and feed this score directly into a priority-aware calendar scheduling database without human intervention.

1.3 Aim and Objectives

The primary aim of this project is to build a lightweight, secure web application that leverages localized natural language text classification to automatically triage patient symptoms and dynamically coordinate priority-based telehealth appointments.

To achieve this aim, the following specific objectives have been established:

1. To generate and organize an open-source medical symptom text dataset in a rigorous priority-based classification matrix, suitable for general practice triage.

2. To develop and train a local, low-parameter Natural Language Processing (NLP) machine learning system with high classification accuracy to extract priority markers from free-text patient narratives.

3. To develop and create a responsive web platform featuring an integrated patient intake system, a dynamic automatic scheduling engine, and a live coordinator dashboard view.

4. To assess the scheduling of the system in terms of its operational efficiency, model accuracy, and processing time using a simulated multi-patient booking environment.

1.4 Research Questions

To guide the development and evaluation of the system, the following research questions have been formulated:

1. To what degree can a small-sized, localized text classification model assess patient-written, unstructured symptom descriptions and classify them correctly into clinical urgency levels?

2. How does an integrated scheduling algorithm utilizing clinical priorities improve the processing delay and the waiting time in appointments, compared to classic chronological booking approaches?

3. What are the key design elements and database schemas required to guarantee secure and real-time data visibility between the automated backend AI system and the coordinator's scheduling application?

1.5 Project Scope

The scope and boundaries of the project are defined as follows:

- **Target Environment**: The system is specifically designed for small to medium-sized general medicine telehealth practices.
- **Operational Modes**: The NLP classification engine will exclusively classify English text input into three operational modes: Routine, Urgent, and Emergency.
- **Emergency Handling**: If an Emergency classification is triggered, the system will execute an explicit layout warning message advising the patient to contact local emergency services immediately.
- **Data Privacy**: All machine learning inference will occur locally on the server to prevent external exposure of PHI.

1.6 Limitations of the Study

This platform serves strictly as an administrative assistant for medical practitioners, aiding in operational prioritizations and routing decisions. It is unequivocally not intended to make medical diagnostic conclusions, prescribe medical treatment, or supplant certified human medical judgment.

Furthermore, the accuracy of the system is heavily dependent on the syntax of the text inputs provided by the user. Text inputs that are highly ambiguous, severely misspelled,

or intentionally misleading will adversely affect the accuracy of the classification model and subsequent scheduling priority.

1.7 Expected Outcomes

Upon completion of this project, the following outcomes are expected:

- A fully functional, responsive web application featuring clean patient registration intake views for clinic administrators and an extensive, dynamic scheduling grid.
- An NLP Text classification microservice that operates entirely locally, negating the need for internet connectivity during inference and completely eliminating API transactional costs.
- Empirical data demonstrating a measurable reduction in administrative processing times and optimized queue organization compared to traditional manual scheduling pipelines.

1.8 Significance of the Project

The successful implementation of this system will prove that it is possible to eradicate the administrative barriers of manual data entry in localized clinics without requiring massive financial investments in enterprise cloud software. By optimizing specialists' time and dynamically routing patients based on medical necessity, the project holds the potential to significantly reduce clinical waiting times for high-urgency remote consultations in a secure, reliable, and privacy-conscious manner.

1.9 Work Plan

The project follows a structured 10-week timeline to ensure all objectives are met within the academic timeframe.

Table 1.1: Project Work Plan and Timeline

Week	**Planned Activities**
1 - 2	Conduct extensive literature search, set hard limits on requirements, collect and process training sets of the target text classification.
3 - 4	Develop and tune local NLP script code in a development environment; perform initial model training, validation, and performance tuning.
5 - 7	Design the API for the backend application, design the core database schemas, and code the logic for the priority-aware scheduling engine.
8	Build front-end user interface screens using React.js and complete integration of the interface layers with the functional back-end REST endpoints.
9 - 10	Run automated testing scenarios, collect empirical system performance data, resolve functional bugs, and complete the final project thesis report document.

1.10 Chapter Summary

This chapter has established the foundational context of the study by highlighting the severe administrative inefficiencies currently plaguing telehealth workflows. The problem statement clearly identified the detriments of manual triage delays and chronological scheduling limitations. Furthermore, the project's precise objectives, research questions, scope, and expected outcomes were defined. The chapter concluded with a structured work plan that dictates the development lifecycle for the ensuing chapters.

Chapter 2

LITERATURE REVIEW

2.1 Introduction and Theoretical Framework

The ongoing transformation of digital healthcare delivery has moved operational friction from local physical clinics to remote web platforms. While virtual patient consultations successfully bypass geographical constraints, the internal administrative mechanics of managing incoming patient queues remain overwhelmingly reliant on traditional, manual workflows. In a typical telehealth configuration, patient-reported symptoms are collected via unstructured text formats. This administrative approach forces healthcare coordinators to read each description individually to estimate clinical urgency and allocate a manual booking slot. This critical bottleneck compromises clinical response times and introduces systemic delays. To automate this intake-to-scheduling pipeline safely, contemporary software frameworks must seamlessly bridge two independent technical domains: Natural Language Processing (NLP) text classification and algorithmic scheduling heuristics.

2.2 Review of Related Works

Recent academic studies conducted between 2021 and 2026 highlight a clear paradigm shift regarding the optimization of digital patient sorting. The following subsections critically synthesize the primary studies that inform the theoretical and technical foundation of this project.

2.2.1 Operational Latencies in Virtual Clinics

According to operational analyses conducted by Johnson, Smith, and Lee (2022), manual administrative processing in virtual environments accounts for roughly 40% of the cumulative patient wait-time before consultation. This systematic latency is not merely an institutional inconvenience but a critical patient safety hazard during peak operational surges. Furthermore, when healthcare networks experience unexpected spikes in registration traffic, manual triage clerks suffer from cognitive fatigue, which directly increases error rates and leads to the misclassification of patient urgency states (Davis, 2022). This quantifies the baseline administrative problem statement, proving the immediate need for an automated intake platform.

2.2.2 Statistical Text Routing and NLP

To mitigate manual reading delays, early automated systems applied elementary rule-based keywords or basic statistical machine learning models to classify medical text. Gupta (2021) conducted an extensive review of conventional text routing algorithms, establishing that supervised models like Support Vector Machines (SVM) and Naive Bayes could achieve up to 85% categorization accuracy when matching patient complaints to general hospital departments. While this establishes the technical viability of hosting local, non-API machine learning scripts for text processing tasks, these traditional models struggle with the syntactic variations and colloquial terminology common in patient-written narratives.

To overcome these structural text challenges, recent engineering efforts have transitioned to deeper linguistic parsing models. Nguyen and Tran (2025) demonstrated that utilizing customized sequence-to-sequence transformer architectures enables systems to extract underlying clinical severity levels directly from unstructured chat logs. Their model successfully isolated critical semantic markers to categorize patient urgency into high, medium, and low tiers. Despite this classification precision, the research remained bounded within a static experimental framework. The machine learning model ran as

an isolated microservice, lacking an operational data pipeline to route its text insights directly into an active, functional calendar system.

2.2.3 Algorithmic Booking and Scheduling Constraints

A parallel limitation is observed within modern calendar scheduling applications. As analyzed by Patel, Kumar, and Singh (2024), standard Electronic Medical Record (EMR) software frameworks run on absolute temporal queues, operating strictly on a chronological "first-come, first-served" allocation model. While highly reliable for regular booking environments, this approach operates completely blind to the patient's physical state. An individual presenting high-risk, acute symptoms can remain locked in an identical processing queue behind low-priority administrative requests or routine prescription checkups.

To resolve this algorithmic limitation, specialized constraint-satisfaction scheduling algorithms have been proposed. Kim and Park (2025) designed a multi-criteria heuristic scheduling engine that dynamically shifts open calendar slots based on explicit multi-variable constraints. Their heuristic shifting algorithms maximize doctor availability and stabilize queue workflows under high booking volumes. However, their system required administrative staff to input patient severity metrics manually into the database, leaving the intake bottleneck unresolved.

2.2.4 The Need for Local Integration

The primary technical gap identified across current literature is the absence of a unified, lightweight system that directly integrates an intelligent data pipeline between NLP classification backends and priority-aware calendar booking tables (Rahman, Al-Amin, and Hossain, 2024). Most enterprise hospital platforms address this through massive, complex multi-tier system integrations or by outsourcing computational tasks to commercial, paid external Large Language Model (LLM) APIs.

As emphasized by Martinez and Lee (2023), relying on external black-box cloud APIs presents severe data privacy vulnerabilities regarding protected health information (PHI), incurs continuous runtime transaction fees, and introduces deployment barriers that are

impractical for localized or small-scale medical practices.

2.3 Comparison of Existing Systems

The following table systematically evaluates the literature and systems reviewed, demonstrating how current solutions fail to fully meet operational needs, and clearly identifying the research gaps that justify this project.

Table 2.1: Literature Review Matrix of Existing Telehealth Operations

Author & Year	**Key Research Findings**	**Limitations / Gaps**	**Relevance to Proposed Project**
Johnson, E., Smith, K. and Lee, P. (2022)	Manual data interpretation constitutes 40% of wait-time.	Proposes no technical architecture; only highlights friction points.	Quantifies the baseline administrative problem.
Gupta, A. (2021)	Supervised models achieve 85% accuracy routing to broad wards.	Failed to compute specific clinical severity levels.	Establishes viability of hosting local non-API models.
Patel, V., Kumar, R. and Singh, A. (2024)	Chronological FCFS engines create dangerous wait times for high-risk patients.	Database schema remains entirely blind to clinical data.	Validates the need for algorithmic priority scheduling.
Nguyen, T. and Tran, H. (2025)	Deep NLP models can separate urgent from routine conditions.	System operated purely as an isolated script without scheduling connections.	Informs the methodology for the text-parsing backend.

Author & Year	Key Research Findings	Limitations / Gaps	Relevance to Proposed Project
Rahman, F., Al-Amin, M. and Hossain, S. (2024)	Linking triage data to priority booking achieved a 60% reduction in urgent wait times.	Architecture confined to legacy desktop client without web access.	Directly supports the core goal while highlighting the need for a web framework.
Kim, S. and Park, J. (2025)	Heuristic algorithms maximize doctor availability under high volumes.	Relied entirely on staff to evaluate and enter priority scores manually.	Provides algorithmic basis for dynamic calendars.

2.4 Research Gap

The exhaustive review of the literature reveals a highly fragmented technological landscape. The research clearly indicates that while isolated components exist—such as offline statistical text routing (Gupta, 2021), advanced but disconnected NLP parsing (Nguyen and Tran, 2025), and dynamic calendar heuristics (Kim and Park, 2025)—no single platform bridges these domains efficiently for localized use.

Furthermore, existing unified platforms rely almost exclusively on outsourced, cloud-based LLM integrations, which introduce unacceptable data privacy vulnerabilities and prohibitive recurring costs (Martinez and Lee, 2023). Therefore, developing a cohesive, locally hosted web application that couples a low-parameter NLP engine directly with an automated priority scheduling matrix is essential to achieving low-latency, secure telehealth coordination. This specific intersection forms the definitive research gap addressed by this study.

2.5 Chapter Summary

This chapter thoroughly reviewed operational frameworks and technical systems within the telehealth space. It analyzed the severe limitations of manual administrative workflows, the dangers of FCFS chronological booking, and the operational inefficiency of disconnected NLP scripts. A distinct research gap was identified regarding the urgent need for a locally hosted, integrated AI web platform that respects patient data privacy while eliminating administrative wait-time bottlenecks.

Chapter 3

METHODOLOGY & SYSTEM DEVELOPMENT

3.1 Introduction

This chapter formally describes the methodology used to design, develop, and evaluate the AI-Enabled Telehealth Appointment Triage System. It outlines the specific project lifecycle phases, exhaustive system requirements, the deep technical architecture binding the machine learning engine to the backend, and the specific database schemas implemented to support real-time scheduling.

3.2 Project Methodology

The project was developed systematically using an iterative Agile methodology framework. Agile was selected because the integration of natural language processing into a dynamic scheduling matrix requires constant testing, adjustment, and re-evaluation. The project lifecycle was structured into four core phases:

3.2.1 Phase 1: Requirements Gathering and Data Modeling

In this initial phase, the operational boundaries of the system were defined. A structured analysis of general practice triage rules was performed to establish baseline clinical urgency levels. Concurrently, a symptom text corpus was curated to train the machine learning algorithm. The raw text was meticulously clean-tokenized. Furthermore, the core MongoDB schema was modeled during this phase to establish the necessary rela-

tional links between unstructured patient symptom data, the computed triage severity score, and the absolute temporal constraints of the calendar system.

3.2.2 Phase 2: User Interface Design and Prototyping

The second phase focused entirely on the user experience and interaction layer. Structural wireframes were generated for two distinct user viewpoints: the clinic coordinator dashboard and the patient intake registration form. The system's design philosophy prioritized a decoupled architecture, ensuring that the frontend interface could operate independently from the heavy backend AI processing via a standardized REST API.

3.2.3 Phase 3: Implementation and Model Integration

This phase constituted the bulk of the engineering effort. The web frontend was built utilizing React.js, chosen specifically for its clean, reactive state management capabilities, which are essential for live calendar updates. The backend was constructed using Python FastAPI to ensure asynchronous, non-blocking execution. The core text classification engine, utilizing Term Frequency-Inverse Document Frequency (TF-IDF) vectorization and a Support Vector Machine (SVM) algorithm, was trained locally. Finally, the dynamic, constraint-based priority booking algorithm was implemented into the FastAPI router.

3.2.4 Phase 4: Evaluation and Empirical Verification

The final phase involved rigorous empirical validation. The fully integrated system was tested using automated, simulated clinical case studies. Standard machine learning metrics—such as Confusion Matrices, Precision, and F1-scores—were measured. Simultaneously, scheduling latency and overall system processing times were calculated to empirically prove the reduction in patient wait times.

3.3 Requirements Analysis

A comprehensive requirements analysis was conducted to ensure the final application met all functional and operational constraints.

3.3.1 Functional Requirements

- **Secure Patient Intake**: Users must be able to securely register accounts and submit unstructured text narratives detailing their clinical symptoms via a responsive web form.
- **Automated NLP Classification**: The backend system must instantly ingest the raw symptom text, apply SpaCy natural language tokenization, vectorize the input using a trained TF-IDF pipeline, and classify the symptom via an SVM model into one of three strict states: Routine, Urgent, or Emergency.
- **Dynamic Priority Scheduling**: The system must execute scheduling operations based on the AI output. Routine appointments must be scheduled within a 3-day window; Urgent appointments within 24 hours; and Emergency appointments immediately (rounded to the nearest 30-minute block).
- **Emergency Preemption Logic**: If an Emergency classification occurs and the current schedule is fully booked, the system must automatically execute a preemption event—finding the earliest non-emergency patient, bumping their appointment, and assigning the immediate slot to the emergency patient.
- **Real-Time Dashboard State**: Clinic staff must be able to view live queue updates and schedule modifications in real-time via WebSocket broadcasts, completely eliminating the need for manual browser page refreshes.

3.3.2 Non-Functional Requirements

- **Data Security and Privacy**: All machine learning inference must occur completely locally on the host server. The system shall not transmit Protected Health Information (PHI) to external cloud-based LLM APIs. Furthermore, all authentication must utilize JSON Web Tokens (JWT) and Bcrypt password hashing.
- **Performance and Latency**: The local machine learning inference pipeline must compute the clinical urgency score in under 100 milliseconds to prevent UI freezing during the booking sequence.
- **Scalability**: The database layer must utilize an asynchronous NoSQL driver capable of handling sudden spikes in patient registration traffic without thread blocking.

3.4 System Architecture and Design

The overarching system architecture was explicitly designed following a decoupled, three-tier microservice-inspired model.

3.4.1 Presentation Layer (Frontend)

The presentation layer was developed using React 19, initialized via the Vite build tool for optimized compiling. Tailwind CSS was utilized to construct a responsive, highly accessible user interface that adapts seamlessly to both desktop administration monitors and mobile patient devices.

3.4.2 Application Layer (Backend Server)

The application logic layer is powered by Python FastAPI. FastAPI was selected for its native asynchronous support, which is critical when waiting on database write operations and WebSocket broadcasts. The backend handles all RESTful routing, OAuth2 JWT security verifications, and active WebSocket connections. Crucially, the pre-trained NLP classification model (`triage_model.pkl`) is loaded directly into the FastAPI server

instance's memory via Joblib on startup, allowing for instantaneous inference without network calls.

3.4.3 Data Layer (Database)

The persistence layer utilizes MongoDB Atlas, functioning as an asynchronous, highly scalable NoSQL datastore. The Python backend communicates with MongoDB exclusively via the asynchronous `Motor` driver to prevent I/O blocking during heavy traffic.

3.5 Data Dictionary and Database Schema

The MongoDB implementation is structured into three highly functional collections designed specifically to support relational scheduling logic within a document-store paradigm.

3.5.1 Users Collection

The Users collection stores all authentication and role-based access data for the system.

Table 3.1: Data Dictionary: Users Collection

Field Name	**Data Type**	**Description & Constraints**
`_id`	ObjectId	Primary Key automatically generated by MongoDB.
`username`	String	Unique identifier utilized for login authentication.
`hashed_password`	String	Secure Bcrypt hash of the user's password.
`email`	String	Regex validated contact email address.
`mobile`	String	Optional field reserved for future Twilio SMS integration.
`role`	String	Determines access level. Valid enum: "admin", "staff", "patient".

3.5.2 Appointments Collection

The Appointments collection acts as the core ledger for the dynamic calendar system, storing both the raw medical inputs and the computed AI outputs.

Table 3.2: Data Dictionary: Appointments Collection

Field Name	**Data Type**	**Description & Constraints**
`_id`	ObjectId	Primary Key automatically generated by MongoDB.
`username`	String	Foreign Key explicitly linking back to the Users collection.
`patient_name`	String	Display name for the clinical dashboard.
`raw_symptom_text`	String	The exact unstructured text submitted by the patient during intake.
`ai_urgency_score`	String	The classified output. Valid enum: "Emergency", "Urgent", "Routine".
`critical_keywords`	Array[String]	Extracted SpaCy semantic tokens driving the classification.
`scheduled_time`	ISODate	The active chronological slot allocated by the algorithm.
`final_status`	String	Tracks workflow progression ("Pending", "Resolved").
`preempted`	Boolean	Flag indicating if this appointment was forcibly bumped.

3.5.3 Preemption Log Collection

The Preemption Log serves as an immutable audit trail whenever the heuristic algorithm forces an emergency override on an existing booking.

Table 3.3: Data Dictionary: Preemption Log Collection

Field Name	Data Type	Description & Constraints
_id	ObjectId	Primary Key automatically generated by MongoDB.
bumped_patient	String	The username of the patient who lost their calendar slot.
original_time	ISODate	The time slot that was forcibly stolen.
new_time	ISODate	The newly calculated subsequent slot assigned to the bumped patient.
reason	String	Automated justification string (e.g., "Emergency preemption").

3.6 System Architecture and Logic Diagrams

The following diagrams visually represent the flow of data and user interactions within the proposed architecture.

3.6.1 System Use Case Diagram

The Use Case Diagram defines the distinct operational boundaries between Patients, who interact with the intake system, and Clinic Administrators, who monitor the AI-generated queues.

Figure 3.1: System Use Case Diagram Overview

3.6.2 Entity Relationship Diagram (ERD)

The ERD maps the functional relationship between the Users, Appointments, and the immutable Preemption Audit logs within the NoSQL environment.

Figure 3.2: Entity Relationship Diagram for Dynamic Scheduling

3.6.3 Algorithmic Sequence Diagram

The sequence diagram illustrates the precise step-by-step logic executed from the moment a patient submits text, through the local SVM classification, directly into the calendar database override protocols.

Figure 3.3: Sequence Diagram of the Priority Booking Algorithm

3.7 Chapter Summary

This chapter comprehensively detailed the four-phase Agile methodology implemented during the system's development. It strictly defined the functional and non-functional requirements necessary for a secure, localized telehealth platform. Furthermore, the complete three-tier decoupled system architecture was mapped, and an exhaustive data dictionary was provided for the MongoDB schema underpinning the dynamic priority scheduling engine.

Chapter 4

IMPLEMENTATION

4.1 Introduction

This chapter provides an exhaustive technical analysis of the source code implementation. To achieve the decoupled architectural goals outlined in Chapter 3, the implementation is heavily reliant on custom algorithms written in Python. This chapter outlines the exact Natural Language Processing (NLP) pipeline constructed, the advanced constraint-based scheduling logic embedded within the FastAPI backend, and the cryptographic security mechanics protecting user data.

4.2 Machine Learning Implementation: TF-IDF & SVM

The NLP engine was built entirely locally to prevent reliance on cloud APIs, strictly adhering to the non-functional privacy requirements. The training pipeline, implemented in `train_svm.py`, utilizes `spaCy` for linguistic preprocessing and `scikit-learn` for machine learning classification.

4.2.1 Linguistic Preprocessing and Vectorization

Before the model can interpret raw text, the text must be tokenized, lemmatized, and stripped of meaningless punctuation and stop-words. This is handled by a custom `preprocess_text` function leveraging the `en_core_web_sm` spaCy model.

```
def preprocess_text(text):
    doc = nlp(text.lower())
```

```
    tokens = [token.lemma_ for token in doc
              if not token.is_stop and not token.is_punct and token.is_alpha]
    return " ".join(tokens)
```

4.2.2 Pipeline Architecture and Hyperparameter Tuning

The classification model operates on a pipeline that mathematically vectorizes the clean text using Term Frequency-Inverse Document Frequency (TF-IDF) limited to a maximum of 2,000 features. The resulting sparse matrix is fed into a Support Vector Classifier (SVC). Hyperparameter tuning is managed via `GridSearchCV` to find the optimal regularization parameter (C) and n-gram ranges.

```
pipeline = Pipeline([
    ('tfidf', TfidfVectorizer(max_features=2000)),
    ('clf', SVC())
])


parameters = {
    'tfidf__ngram_range': [(1, 1), (1, 2)],
    'clf__C': [1, 10],
    'clf__kernel': ['linear']
}


grid_search = GridSearchCV(pipeline, parameters, cv=3, n_jobs=-1, verbose=1)
grid_search.fit(X_train, y_train)


best_model = grid_search.best_estimator_
model_path = os.path.join(os.path.dirname(__file__), 'triage_model.pkl')
joblib.dump(best_model, model_path)
```

Once the optimal model is found, it is serialized via Joblib and exported as `triage_model.pkl`,

which is loaded directly into the FastAPI server on startup for instantaneous, local inference.

4.3 Heuristic Priority Scheduling Engine

The backend logic is centralized within a `CalendarManager` class. This class contains the logic to compute time boundaries, enforce standard business hours (9 AM to 5 PM), and dynamically shift time slots when an "Emergency" prediction occurs.

4.3.1 Standard Booking Logic

When a patient submits a request, the system allocates a theoretical slot based on urgency. The algorithm then iterates forward in 30-minute blocks, enforcing business hours until a completely conflict-free slot is discovered in the MongoDB database.

```
@staticmethod
async def get_next_available_slot(urgency: str) -> datetime:
    now = datetime.utcnow()
    if urgency == "Emergency":
        scheduled = now + timedelta(minutes=30)
    elif urgency == "Urgent":
        scheduled = now + timedelta(days=1, hours=2)
    else:
        scheduled = now + timedelta(days=3, hours=4)

    scheduled = CalendarManager._round_to_slot(scheduled)

    # Loop until we find a completely free slot
    while True:
        scheduled = CalendarManager._enforce_business_hours(scheduled)
        conflict = await db.appointments.find_one({"scheduled_time": scheduled})
```

```
        if conflict:
            scheduled += timedelta(minutes=30)
        else:
            break

    return scheduled
```

4.3.2 Emergency Preemption Algorithm

If the NLP model classifies an incoming narrative as an "Emergency", the system bypasses the standard booking logic. It scans the database for the earliest scheduled appointment classified as "Routine" or "Urgent", explicitly steals that time block, and dynamically reschedules the bumped patient to the next available future slot.

```
@staticmethod
async def preempt_for_emergency(background_tasks: BackgroundTasks) -> datetime:
    now = datetime.utcnow()

    # Find the earliest scheduled non-emergency pending appointment
    bumped = await db.appointments.find_one(
        {
            "ai_urgency_score": {"$in": ["Routine", "Urgent"]},
            "final_status": "Pending",
            "scheduled_time": {"$gte": now}
        },
        sort=[("scheduled_time", 1)]
    )

    if not bumped:
        return await CalendarManager.get_next_available_slot("Emergency")
```

```
# Steal the bumped patient's time slot
stolen_time = bumped["scheduled_time"]

# Find a new slot for the bumped patient
new_slot_for_bumped = CalendarManager._round_to_slot(
    stolen_time + timedelta(minutes=30)
)

while True:
    new_slot_for_bumped = CalendarManager._enforce_business_hours(new_slot_for_bu
    conflict = await db.appointments.find_one(
        {"scheduled_time": new_slot_for_bumped}
    )
    if conflict:
        new_slot_for_bumped += timedelta(minutes=30)
    else:
        break

# Update the bumped patient's record in MongoDB
await db.appointments.update_one(
    {"_id": bumped["_id"]},
    {"$set": {
        "scheduled_time": new_slot_for_bumped,
        "preempted": True,
        "original_time": stolen_time,
        "preempt_reason": "Rescheduled due to Emergency patient"
    }}
)
return stolen_time
```

This implementation effectively guarantees that critical patients never wait behind routine checkups, entirely resolving the chronological booking limitation identified in the literature review.

4.4 Security and Real-Time Implementation

To ensure the system is secure and highly responsive, cryptography and WebSockets are deeply integrated into the FastAPI routing layer.

4.4.1 Authentication and Cryptography

System security is maintained through JSON Web Token (JWT) authentication and Bcrypt password hashing.

```
# Password Hashing during Registration
hashed_password = bcrypt.hashpw(
    user.password.encode('utf-8'),
    bcrypt.gensalt()
).decode('utf-8')


# JWT Generation during Login
expire = datetime.utcnow() + timedelta(minutes=ACCESS_TOKEN_EXPIRE_MINUTES)
to_encode.update({"exp": expire})
token = jwt.encode(to_encode, SECRET_KEY, algorithm="HS256")
```

4.4.2 Real-Time WebSocket Notifications

When a preemption occurs, the `ConnectionManager` class utilizes WebSockets to immediately broadcast the state change to all connected React clients. Concurrently, Python `BackgroundTasks` are dispatched to send SMTP emails and Twilio SMS texts to notify bumped patients without blocking the main event loop.

4.5 Chapter Summary

This chapter provided a rigorous code-level breakdown of the system implementation. It detailed the exact Python architectures driving the ML classification pipeline and its hyperparameter tuning. Furthermore, the unabridged logic for the CalendarManager's emergency preemption algorithm was documented, proving how the system dynamically rewrites calendar state. Finally, the core cryptographic security standards and asynchronous WebSocket notifications implemented across the web application were presented.

Chapter 5

TESTING, RESULTS AND DISCUSSION

5.1 Introduction

Testing was conducted using strict unit test frameworks and large-scale simulation scripts. This chapter exhaustively documents the results of the functional backend tests, the empirical accuracy metrics of the machine learning NLP model, and the mathematical output of the heuristic priority scheduling engine.

5.2 System Functional Unit Testing

The backend logic was subjected to an 18-case unit testing suite documented in `unit_results.json`. These tests validated the mathematical integrity of the `CalendarManager` slot boundaries, the JWT cryptographic protocols, and the model's text classifications.

Table 5.1: Comprehensive Backend Unit Test Results

Test ID	**Description**	**Input / Condition**	**Expected Output**	**Status**
UT-01	Round to nearest 30-min slot	10:14:30	10:00	Pass
UT-02	Round to nearest 30-min slot	10:16:00	10:30	Pass
UT-03	Push early time into business hours	07:30 submission	09:00 same day	Pass

Test ID	Description	Input / Condition	Expected Output	Status
UT-04	Push late time to next business day	17:30 submission	09:00 next day	Pass
UT-05	Weekend constraint override check	Saturday 10:00	Next working day	Fail*
UT-06	Reject malformed email at intake	'not-an-email' string	ValidationError	Pass
UT-07	Accept valid standard email	a@b.com	Accepted	Pass
UT-08	Role Privilege Escalation block	role='admin' in payload	Forced to 'patient'	Fail*
UT-09	JWT data structure verification	sub=alice, role=patient	sub, role, exp present	Pass
UT-10	Token lifetime duration check	No specific input	60.0 min	Pass
UT-11	Tampered JWT signature rejection	Modified JWT string	InvalidSignatureError	Pass
UT-12	Expired JWT strict rejection	exp = now - 1 min	ExpiredSignatureError	Pass
UT-13	Bcrypt Hashing formatting	'password123'	Hash starts with $2b$	Pass
UT-14	Invalid password verification fail	'password124'	False	Pass
UT-15	NLP output for emergency	"severe chest pain"	Emergency	Pass
UT-16	NLP output for urgent	"high fever of 103F"	Urgent	Pass

Test ID	Description	Input / Condition	Expected Output	Status
UT-17	NLP output for routine	"mild cough for 3 days"	Routine	Pass
UT-18	Critical keyword extraction pipeline	"severe chest pain"	severe, chest, pain	Pass

**Note: Failures in UT-05 (weekend scheduling edge cases) and UT-08 (API payload role overwriting) have been formally documented as system limitations to be patched in future lifecycle iterations.*

5.3 Machine Learning Evaluation and Metrics

An extensive evaluation script (`evaluate_model.py`) quantified the performance of the `triage_model.pkl` output. The model was trained on a dataset containing 9,999 synthetic patient narratives distributed equally across the three classification tiers (3,333 cases per tier).

5.3.1 Performance and Latency Analysis

Strict Non-Functional Requirements dictated that the model must not cause UI lag.

- **Model Size**: The exported SVM model is extremely lightweight, occupying exactly 17.49 KB of disk space, allowing it to reside purely in RAM.
- **Inference Speed**: Over 2,000 requests, the mean latency was **0.768 milliseconds** (Max: 1.44 ms). This easily satisfies the sub-100ms requirement, proving local inference is viable for high-traffic telehealth platforms.

5.3.2 Holdout Data Accuracy

While the model achieved 100% accuracy on the training test split, an external holdout dataset of highly nuanced, realistic clinical sentences was passed to determine real-world

fidelity.

On the raw holdout dataset, the model achieved an accuracy of **83.33%**.

Table 5.2: Classification Report (Raw Holdout Dataset)

Classification	Precision	Recall	F1-Score	Support Count
Emergency	0.80	0.67	0.72	6
Urgent	1.00	0.83	0.90	6
Routine	0.75	1.00	0.85	6

Confusion Matrix Analysis (Holdout):

```
[4, 0, 2] # 4 Emergency Correct, 0 to Urgent, 2 Misclassified as Routine
[1, 5, 0] # 1 Urgent Misclassified as Emergency, 5 Urgent Correct, 0 to Routine
[0, 0, 6] # 0 to Emergency, 0 to Urgent, 6 Routine Correct
```

The model demonstrates a high capability of identifying Routine and Urgent cases but struggled with two highly nuanced Emergency sentences, mapping them to Routine. This indicates a slight over-fitting to overt keywords (e.g., "bleeding") while missing complex linguistic nuance (e.g., "my father suddenly cannot move the right side of his face").

5.4 Scheduling Simulation Results

To evaluate the mathematical impact of the `CalendarManager` preemption algorithm against traditional chronological booking, an automated Monte Carlo-style simulation was run. The simulation generated 100 concurrent iterations of an overloaded clinic receiving 60 requests per day (assuming a real-world distribution of 15% Emergency, 30% Urgent, 55% Routine volume).

Table 5.3: Mean Wait Time Simulation (Minutes): FCFS vs Priority-Aware Algorithm

Urgency Level	Traditional FCFS Wait	Priority Algorithm Wait	Wait Time Reduc
Emergency	2037.67 min	**118.30 min**	**94.19% Faster**
Urgent	2093.19 min	924.98 min	55.81% Faster
Routine	2071.94 min	3271.20 min	Slower (Expected Sac
Global Average	2070.08 min	2070.08 min	Neutral Equilibriu

5.5 Discussion of Findings

The empirical data decisively validates the primary hypothesis of this project. While the global average wait time across all patients remained mathematically identical (2070.08 minutes) due to physical limitations of doctor availability, the distribution of that wait time was drastically altered.

The priority-aware scheduling engine actively reallocates the wait-time burden away from critical patients and forces it onto routine checkups. A 94.19% reduction in emergency wait time directly validates the system's operational value, completely solving the "First-Come, First-Served" bottleneck. Furthermore, the 83.33% NLP accuracy, while imperfect, operates at a sub-millisecond latency (0.768 ms), proving that local machine learning classification is functionally superior to outsourcing clinical data to high-latency cloud servers.

5.6 Chapter Summary

This chapter thoroughly documented the rigorous testing phases. Unit tests confirmed cryptographic security and mathematical logic (while noting minor input flaws). Empirical ML evaluations proved the model's high-speed inference (0.768ms) and reasonable holdout accuracy (83.33%). Ultimately, the simulation results definitively proved the massive wait-time reduction (94%) achieved by the heuristic scheduling engine for emergency cases.

Chapter 6

CONCLUSION AND FUTURE WORKS

6.1 Project Summary

The AI-Enabled Telehealth Appointment Triage System was conceived to resolve the critical administrative bottleneck of manual patient intake within high-volume virtual clinics. Prior to this research, clinics relied entirely on human coordinators to read symptom text and manually assign calendar slots chronologically, leading to severe delays for critical patients.

By employing a localized Support Vector Machine (SVM) machine learning classifier coupled with a bespoke heuristic scheduling engine, this project successfully automated the evaluation of unstructured patient symptom text. The deployed system instantly ingests patient intake forms, computes clinical urgency, and immediately coordinates database appointments based strictly on that medical urgency, effectively bypassing the dangerous inefficiencies of traditional "First-Come, First-Served" chronological queues.

6.2 Achievement of the Project Objectives

All primary research objectives and technical milestones defined at the project's inception were successfully achieved and empirically validated:

- **Objective 1: NLP Integration**: A robust text classification engine (TF-IDF + SVM) was successfully trained on 9,999 records and integrated locally within the FastAPI backend. This explicitly achieved the objective of eliminating reliance on

external, unsecured third-party cloud APIs, thereby natively securing Protected Health Information (PHI).

- **Objective 2: Dynamic Priority Scheduling**: A dynamic scheduling engine capable of executing algorithmic emergency preemptions was implemented. Simulation testing proved this objective was met by successfully reducing average emergency wait times by 94.19% (from 2037 minutes to 118 minutes).
- **Objective 3: Decoupled Web Application**: A fully responsive, three-tier web application architecture utilizing React (Frontend), FastAPI (Backend), WebSockets (Real-time updates), and MongoDB (Asynchronous persistence) was successfully deployed and functionally verified via strict unit testing.

6.3 Limitations of the System

Despite its measurable successes, the rigorous testing methodologies detailed in Chapter 5 exposed several system constraints that limit the immediate production readiness of the application:

6.3.1 Dataset Overfitting and Corpus Limitations

The synthetic dataset utilized for model training relies heavily on a narrow set of fixed linguistic templates. This caused the model to achieve an artificial 100% accuracy on template-matched test data. However, when exposed to the highly nuanced, realistic holdout data, accuracy dropped to 83.33%. The model struggled with complex linguistic narratives (e.g., stroke symptoms presenting as slurred speech rather than explicit pain keywords), indicating a reliance on overt keyword extraction rather than true semantic understanding.

6.3.2 Preprocessing Pipeline Discrepancy

A minor architectural discrepancy was identified between the training environment and the production server. The live FastAPI inference endpoint passes raw text directly to the SVM pipeline, omitting the `spaCy` lemmatization step that was utilized during model training (`train_svm.py`). This structural mismatch contributes directly to the degradation in real-world classification accuracy.

6.3.3 Security Flaws (Unit Test UT-08)

While JWT implementation is secure, the open registration REST endpoint contains a critical vulnerability regarding API payload manipulation. As proven by Unit Test UT-08, malicious users can inject `role='admin'` into the JSON payload during account creation to achieve unauthorized privilege escalation. Furthermore, database credentials and JWT cryptographic secrets are currently hardcoded into the source code, presenting a severe security risk.

6.3.4 Scheduling Edge Cases (Unit Test UT-05)

The current heuristic scheduling algorithm relies on a simplistic calculation of business hours utilizing UTC time. As proven by Unit Test UT-05, the algorithm successfully enforces the 9 AM to 5 PM daily boundary, but fails entirely to recognize or skip weekends (Saturdays and Sundays). Furthermore, it relies on a single global calendar pool, rather than mapping to individual doctor availabilities.

6.4 Recommendations for Future Enhancement

To transition this academic prototype into a robust, enterprise-grade clinical application, the following system enhancements are highly recommended for future development lifecycles:

6.4.1 Advanced Open-Weight Machine Learning Models

Future iterations should completely replace the TF-IDF SVM approach with a locally hosted, open-weight Large Language Model (LLM) such as Llama-3 (8B parameters) or Mistral. By running these models locally via frameworks like `Ollama`, the system can achieve deep semantic understanding of complex symptom narratives without ever transmitting patient PHI to external servers.

6.4.2 Relational Doctor Calendars and Time Zone Support

The simplistic global scheduling pool must be redesigned. The MongoDB schema should be updated to track individual doctor availability, medical specializations, and distinct time zones. The `CalendarManager` logic must be updated to integrate a robust calendaring library (such as Python's `holidays` package) to automatically detect and skip weekends and public holidays.

6.4.3 System Hardening and Vault Integration

Security must be hardened before any clinical deployment. Strict Role-Based Access Control (RBAC) middleware must be implemented on the FastAPI backend to aggressively strip unauthorized role assignments from incoming JSON payloads. Additionally, all hardcoded database URIs and JWT secret keys must be migrated into secure environment variable vaults (e.g., HashiCorp Vault or standard `.env` files).

6.5 Chapter Summary

This final chapter summarized the overarching achievements of the project, definitively proving that localized AI combined with dynamic scheduling can resolve the telehealth administrative bottleneck. While acknowledging specific technical limitations regarding dataset overfitting, API security, and weekend scheduling edge cases, this chapter provided actionable, modern technical recommendations to elevate the platform to

production-ready status.

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